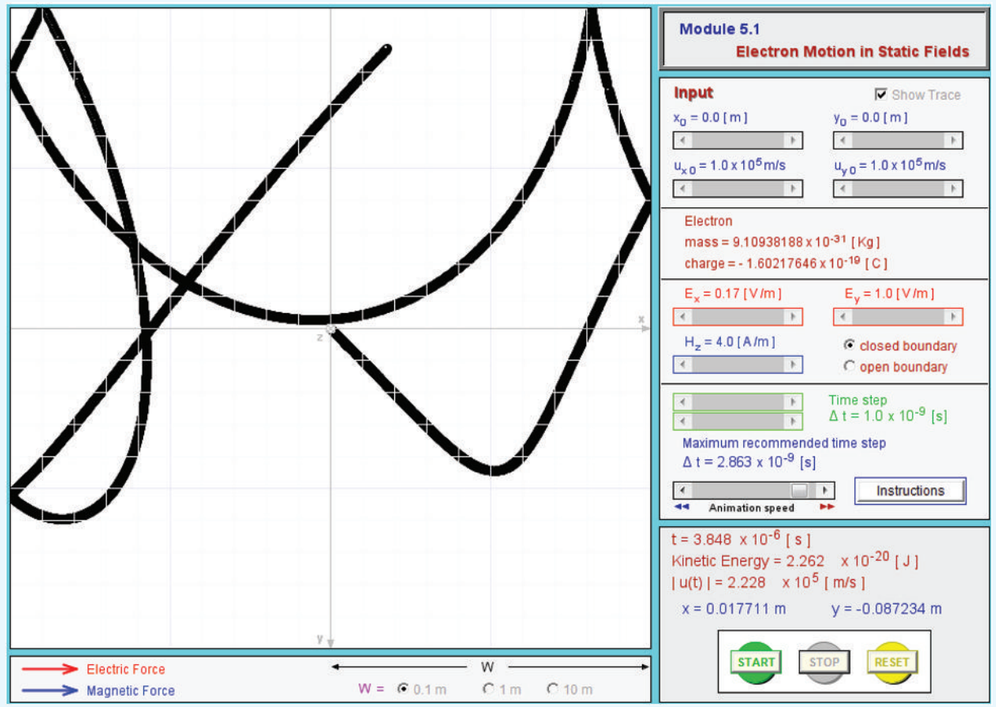
Module 5.1 Electron Motion in Static Fields This module demonstrates the Lorentz force on an electron moving under the influence of an electric field alone, a magnetic field alone, or both acting simultaneously.



If the closed wire shown in **Fig. 5-3(a)** resides in a uniform external magnetic field **B**, then **B** can be taken outside the integral in Eq. (5.10), in which case

$$\mathbf{F}_m = I\left(\oint_C d\mathbf{l}\right) \times \mathbf{B} = 0. \tag{5.11}$$

► This result, which is a consequence of the fact that the vector sum of the infinitesimal vectors $d\mathbf{l}$ over a closed path equals zero, states that the total magnetic force on any closed current loop in a uniform magnetic field is zero. ◄

This does not mean that the force at every point along the wire is zero, but rather that the vector sum of all forces exerted on all parts of the closed wire adds up to zero.

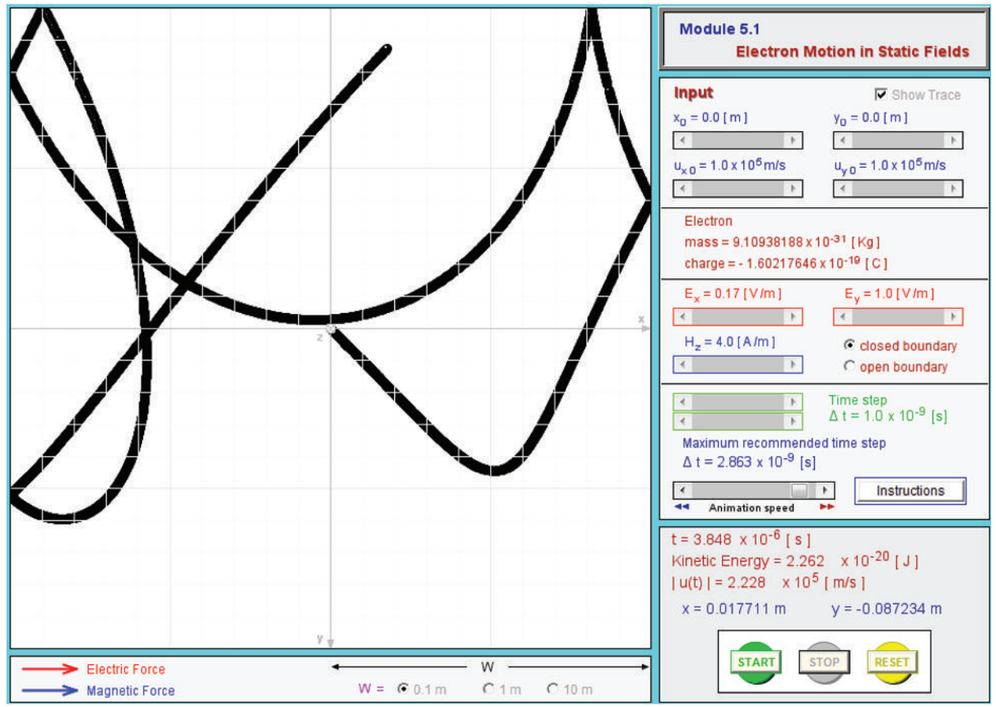
In the study of magnetostatics, all currents flow through closed paths. To understand why, consider the curved wire in **Fig. 5-3(b)** carrying a current I from point a to point b. In doing so, negative charges accumulate at a, and positive ones at b. The time-varying nature of these charges violates the static assumptions underlying Eqs. (5.1a,b).

If we are interested in the magnetic force exerted on a wire segment l (**Fig. 5-3(b)**) residing in a uniform magnetic field (while realizing that it is part of a closed current loop), we can integrate Eq. (5.9) to obtain

$$\mathbf{F}_m = I\left(\int_\ell d\mathbf{l}\right) \times \mathbf{B} = I\boldsymbol{\ell} \times \mathbf{B}, \tag{5.12}$$

where $\boldsymbol{\ell}$ is the vector directed from a to b (**Fig. 5-3(b)**). The integral of $d\mathbf{l}$ from a to b has the same value irrespective of the path taken between a and b. For a closed loop, points a and b

模块 5.1 静态场中的电子运动 该模块演示了在单独的电场、单独的磁场或两者同时作用的影响下运动的电子所受到的洛伦兹力。



如果图 5-3(a) 所示的闭合导线位于均匀的外部磁场 B 中，则 B 可以在方程 (1) 的积分之外取。 (5.10)，在这种情况下

$$\mathbf{F}_m = I\left(\oint_C d\mathbf{l}\right) \times \mathbf{B} = 0. \tag{5.11}$$

► 这个结果是闭合路径上无穷小矢量 dl 的矢量和等于零这一事实的结果，表明均匀磁场中任何闭合电流环路上的总磁力为零。 ◄

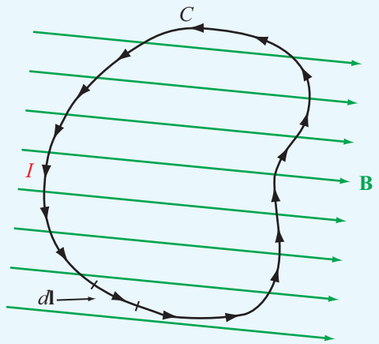
这并不意味着沿线的每一点的力都为零，而是施加在闭合线的所有部分上的所有力的矢量和加起来为零。

在静磁学的研究中，所有电流都流过闭合路径。要理解其中的原因，请考虑图 5-3(b) 中的弯曲导线，其从 a 点到 b 点承载电流 I。在此过程中，负电荷在 a 处积聚，正电荷在 b 处积聚。这些电荷的随时间变化的性质违反了方程式的静态假设。 (5.1a，b) 。

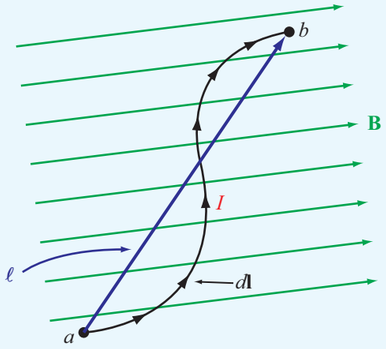
如果我们对施加在均匀磁场中的线段 l（图 5-3(b)）上的磁力感兴趣（同时意识到它是闭合电流环路的一部分），我们可以对方程 1 进行积分。 (5.9) 得到

$$\mathbf{F}_m = I\left(\int_\ell d\mathbf{l}\right) \times \mathbf{B} = I\boldsymbol{\ell} \times \mathbf{B}, \tag{5.12}$$

其中$\boldsymbol{\ell}$是从a到b的向量（图5-3（b））。无论 a 和 b 之间采用的路径如何，从 a 到 b 的 dl 积分都具有相同的值。对于闭环，点 a 和 b



(a)



(b)

Figure 5-3 In a uniform magnetic field, (a) the net force on a closed current loop is zero because the integral of the displacement vector $d\mathbf{l}$ over a closed contour is zero and (b) the force on a line segment is proportional to the vector between the end point ($\mathbf{F}_m = I\boldsymbol{\ell} \times \mathbf{B}$).

become the same point, in which case $\boldsymbol{\ell} = 0$ and $\mathbf{F}_m = 0$.

Example 5-1: Force on a Semicircular Conductor

The semicircular conductor shown in **Fig. 5-4** lies in the

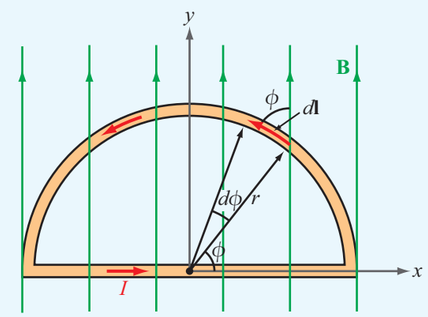


Figure 5-4 Semicircular conductor in a uniform field (Example 5-1).

x–y plane and carries a current I. The closed circuit is exposed to a uniform magnetic field $\mathbf{B} = \hat{\mathbf{y}}B_0$. Determine (a) the magnetic force $\mathbf{F}_1$ on the straight section of the wire and (b) the force $\mathbf{F}_2$ on the curved section.

Solution: (a) To evaluate $\mathbf{F}_1$, consider that the straight section of the circuit is of length $2r$ and its current flows along the $+x$ direction. Application of Eq. (5.12) with $\boldsymbol{\ell} = \hat{\mathbf{x}}\,2r$ gives

$$\mathbf{F}_1 = \hat{\mathbf{x}}(2Ir) \times \hat{\mathbf{y}}B_0 = \hat{\mathbf{z}}\,2IrB_0 \qquad \text{(N)}.$$

The $\hat{\mathbf{z}}$ direction in **Fig. 5-4** is *out of the page*.

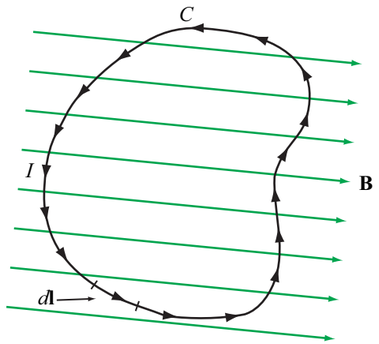
(b) To evaluate $\mathbf{F}_2$, consider a segment of differential length $d\mathbf{l}$ on the curved part of the circle. The direction of $d\mathbf{l}$ is chosen to coincide with the direction of the current. Since $d\mathbf{l}$ and $\mathbf{B}$ are both in the x–y plane, their cross product $d\mathbf{l} \times \mathbf{B}$ points in the negative z direction, and the magnitude of $d\mathbf{l} \times \mathbf{B}$ is proportional to $\sin\phi$, where ϕ is the angle between $d\mathbf{l}$ and $\mathbf{B}$. Moreover, the magnitude of $d\mathbf{l}$ is $dl = r\,d\phi$. Hence,

$$\mathbf{F}_2 = I\int_{\phi=0}^{\pi} d\mathbf{l} \times \mathbf{B}$$

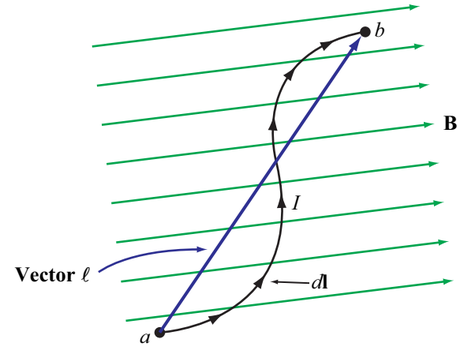
$$= -\hat{\mathbf{z}}I\int_{\phi=0}^{\pi} rB_0 \sin\phi\, d\phi = -\hat{\mathbf{z}}\,2IrB_0 \qquad \text{(N)}.$$

The $-\hat{\mathbf{z}}$ direction of the force acting on the curved part of the conductor is *into the page*. We note that $\mathbf{F}_2 = -\mathbf{F}_1$, implying that no net force acts on the closed loop, although opposing forces act on its two sections.

Concept Question 5-1: What are the major differences between the electric force $\mathbf{F}_e$ and the magnetic force $\mathbf{F}_m$?



(a)



(b)

图 5-3 在均匀磁场中，(a) 闭合电流环路上的合力为零，因为闭合轮廓上的位移矢量 dl 的积分为零；(b) 线段上的力与端点 ($F_m = I\ell \times B$) 之间的矢量成正比。

变为同一点，在这种情况下 $\ell = 0$ 和 $F_m = 0$。

例 5-1：半圆形导体上的力

图5-4所示的半圆形导体位于

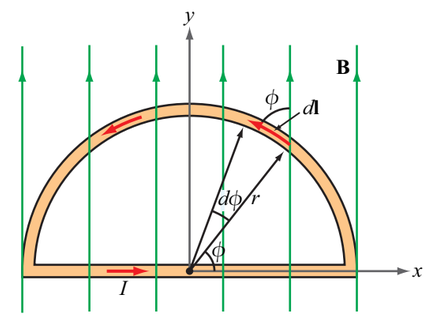


图 5-4 均匀场中的半圆形导体（例 5-1）。

x–y 平面并承载电流 I。闭合电路暴露在均匀磁场 $B = \hat{y}B_0$ 中。确定 (a) 导线直线部分上的磁力 F_1 和 (b) 弯曲部分上的力 F_2。

解：(a) 要计算 F_1，请考虑电路的直线部分长度为 2r，并且其电流沿 +x 方向流动。方程的应用(5.12) 与 $\ell = \hat{x}\,2r$ 给出

$$\mathbf{F}_1 = \hat{\mathbf{x}}(2Ir) \times \hat{\mathbf{y}}B_0 = \hat{\mathbf{z}}\,2IrB_0 \qquad \text{(N)}.$$

图5-4中的$\hat{z}$方向在页面之外。

(b) 为了计算 F_2，请考虑圆的弯曲部分上的微分长度 dl 的线段。选择 dl 的方向与电流方向一致。由于 dl 和 B 都在 x-y 平面上，因此它们的叉积 dl × B 指向负 z 方向，并且 dl × B 的大小与 sin ϕ 成正比，其中 ϕ 是 dl 和 B 之间的角度。此外，dl 的大小为 $dl = r\,d\phi$。因此，

$$\mathbf{F}_2 = I\int_{\phi=0}^{\pi} d\mathbf{l} \times \mathbf{B}$$

$$= -\hat{\mathbf{z}}I\int_{\phi=0}^{\pi} rB_0 \sin\phi\, d\phi = -\hat{\mathbf{z}}\,2IrB_0 \qquad \text{(N)}.$$

作用在导体弯曲部分的力的$-\hat{z}$方向进入页面。我们注意到 $F_2 = -F_1$，这意味着没有净力作用在闭环上，尽管相反的力作用在其两个部分上。

概念问题 5-1：电力 F_e 和磁力 F_m 之间的主要区别是什么？

Concept Question 5-2: The ends of a 10-cm long wire carrying a constant current I are anchored at two points on the x axis, namely $x = 0$ and $x = 6$ cm. If the wire lies in the x–y plane in a magnetic field $\mathbf{B} = \hat{\mathbf{y}}B_0$, which of the following arrangements produces a greater magnetic force on the wire: (a) wire is V-shaped with corners at $(0,0)$, $(3,4)$, and $(6,0)$ or (b) wire is an open rectangle with corners at $(0,0)$, $(0,2)$, $(6,2)$, and $(6,0)$?

Exercise 5-4: A horizontal wire with a mass per unit length of 0.2 kg/m carries a current of 4 A in the $+x$ direction. If the wire is placed in a uniform magnetic flux density $\mathbf{B}$, what should the direction and minimum magnitude of $\mathbf{B}$ be in order to magnetically lift the wire vertically upward? (*Hint:* The acceleration due to gravity is $\mathbf{g} = -\hat{\mathbf{z}}9.8$ m/s^2.)

Answer: $\mathbf{B} = \hat{\mathbf{y}}0.49$ T. (See EM.)

5-1.2 Magnetic Torque on a Current-Carrying Loop

When a force is applied on a rigid body that can pivot about a fixed axis, the body will, in general, react by rotating about that axis. The angular acceleration depends on the cross product of the applied force vector $\mathbf{F}$ and the distance vector $\mathbf{d}$, which is measured *from* a point on the rotation axis (such that $\mathbf{d}$ is perpendicular to the axis) *to* the point of application of $\mathbf{F}$ (Fig. 5-5). The length of $\mathbf{d}$ is called the ***moment arm***, and the cross product

$$\mathbf{T} = \mathbf{d} \times \mathbf{F} \qquad \text{(N·m)} \tag{5.13}$$

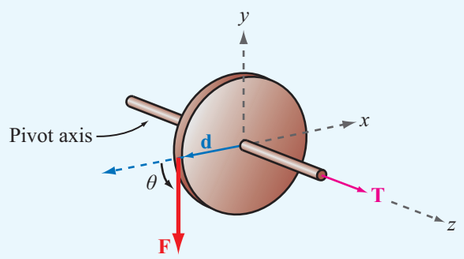


Figure 5-5 The force $\mathbf{F}$ acting on a circular disk that can pivot along the z axis generates a torque $\mathbf{T} = \mathbf{d} \times \mathbf{F}$ that causes the disk to rotate.

is called the ***torque***. The unit for $\mathbf{T}$ is the same as that for work or energy, even though torque does not represent either. The force $\mathbf{F}$ applied on the disk shown in Fig. 5-5 lies in the x–y plane and makes an angle θ with $\mathbf{d}$. Hence,

$$\mathbf{T} = \hat{\mathbf{z}}rF \sin\theta, \tag{5.14}$$

where $|\mathbf{d}| = r$, which is the radius of the disk, and $F = |\mathbf{F}|$. From Eq. (5.14), we observe that a torque along the positive z direction corresponds to a tendency for the cylinder to rotate counterclockwise and, conversely, a torque along the $-z$ direction corresponds to clockwise rotation.

► These directions are governed by the following ***right-hand rule:*** When the thumb of the right hand points along the direction of the torque, the four fingers indicate the direction that the torque tries to rotate the body. ◄

We now consider the ***magnetic torque*** exerted on a conducting loop under the influence of magnetic forces. We begin with the simple case where the magnetic field $\mathbf{B}$ is in the plane of the loop, and then we extend the analysis to the more general case where $\mathbf{B}$ makes an angle θ with the surface normal of the loop.

Magnetic Field in the Plane of the Loop

The rectangular conducting loop shown in Fig. 5-6(a) is constructed from rigid wire and carries a current I. The loop lies in the x–y plane and is allowed to pivot about the axis shown. Under the influence of an externally generated uniform magnetic field $\mathbf{B} = \hat{\mathbf{x}}B_0$, arms 1 and 3 of the loop are subjected to forces $\mathbf{F}_1$ and $\mathbf{F}_3$ given by

$$\mathbf{F}_1 = I(-\hat{\mathbf{y}}b) \times (\hat{\mathbf{x}}B_0) = \hat{\mathbf{z}}IbB_0, \tag{5.15a}$$

and

$$\mathbf{F}_3 = I(\hat{\mathbf{y}}b) \times (\hat{\mathbf{x}}B_0) = -\hat{\mathbf{z}}IbB_0. \tag{5.15b}$$

These results are based on the application of Eq. (5.12). We note that the magnetic forces acting on arms 1 and 3 are in opposite directions, and no magnetic force is exerted on either arm 2 or 4 because $\mathbf{B}$ is parallel to the direction of the current flowing in those arms.

A bottom view of the loop, depicted in Fig. 5-6(b), reveals that forces $\mathbf{F}_1$ and $\mathbf{F}_3$ produce a torque about the origin O, causing the loop to rotate in a clockwise direction. The moment arm is $a/2$ for both forces, but $\mathbf{d}_1$ and $\mathbf{d}_3$ are in opposite directions, resulting in a total magnetic torque of

$$\begin{aligned}\mathbf{T} &= \mathbf{d}_1 \times \mathbf{F}_1 + \mathbf{d}_3 \times \mathbf{F}_3 \\ &= \left(-\hat{\mathbf{x}}\frac{a}{2}\right) \times (\hat{\mathbf{z}}IbB_0) + \left(\hat{\mathbf{x}}\frac{a}{2}\right) \times (-\hat{\mathbf{z}}IbB_0) \\ &= \hat{\mathbf{y}}IabB_0 = \hat{\mathbf{y}}IAB_0,\end{aligned} \tag{5.16}$$

概念问题 5-2：承载恒定电流 I 的 10 厘米长导线的末端固定在 x 轴上的两个点，即 x = 0 和 x = 6 厘米。如果导线位于磁场 B = ŷB₀ 中的 x-y 平面上，下列哪种布置会在导线上产生更大的磁力：(a) 导线为 V 形，角点为 (0, 0)、(3, 4) 和 (6, 0) 或 (b) 导线为开放矩形，角点为 (0, 0)、(0, 2)、(6, 2) 和 (6, 0) ？

练习 5-4：单位长度质量为 0.2 kg/m 的水平导线在 + x 方向承载 4 A 的电流。如果将导线放置在均匀磁通密度 B 中，为了使导线垂直向上磁力提升，B 的方向和最小大小应为多少？ （提示：重力加速度为 g = −ẑ9.8 m/s²。）

答案：B = ŷ0.49 T。（参见EM。）

5-1.2 载流回路上的磁扭矩

当一个力施加在一个可以绕固定轴旋转的刚体上时，该刚体通常会通过绕该轴旋转来做出反应。角加速度取决于施加的力矢量 F 和距离矢量 d 的叉积，距离矢量 d 是从旋转轴上的一点（使得 d 垂直于轴）到 F 的施加点测量的（图 5-5）。 d 的长度称为力臂，叉积

$$\mathbf{T} = \mathbf{d} \times \mathbf{F} \qquad \text{(N·m)} \tag{5.13}$$

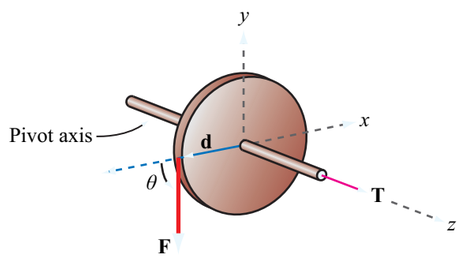


图 5-5 作用在可沿 z 轴枢转的圆盘上的力 F 会产生导致圆盘旋转的扭矩 T = d×F。

称为扭矩。 T 的单位与功或能量的单位相同，尽管扭矩并不代表这两者。如图 5-5 所示，施加在圆盘上的力 F 位于 x-y 平面内，并与 d 形成角度 θ。因此，

$$\mathbf{T} = \hat{\mathbf{z}}rF \sin\theta, \tag{5.14}$$

其中 |d| = r 是圆盘的半径，F = |F|。从方程。根据方程（5.14），我们观察到沿正 z 方向的扭矩对应于圆柱体逆时针旋转的趋势，相反，沿 −z 方向的扭矩对应于顺时针旋转。

► 这些方向受以下右手法则控制：当右手的拇指指向扭矩的方向时，四个手指指示扭矩试图旋转物体的方向。 ◄

我们现在考虑在磁力影响下施加在导电环上的磁扭矩。我们从磁场 B 位于环平面内的简单情况开始，然后将分析扩展到更一般的情况，其中 B 与环的表面法线形成角度 θ。

环平面内的磁场

图 5-6(a) 所示的矩形导电环由刚性导线构成，承载电流 I。该环位于 x-y 平面内，可以绕所示轴旋转。在外部产生的均匀磁场 B = x̂B₀ 的影响下，环路的臂 1 和 3 受到由下式给出的力 F₁ 和 F₃

$$\mathbf{F}_1 = I(-\hat{\mathbf{y}}b) \times (\hat{\mathbf{x}}B_0) = \hat{\mathbf{z}}IbB_0, \tag{5.15a}$$

and

$$\mathbf{F}_3 = I(\hat{\mathbf{y}}b) \times (\hat{\mathbf{x}}B_0) = -\hat{\mathbf{z}}IbB_0. \tag{5.15b}$$

这些结果基于方程的应用。 （5.12）。我们注意到，作用在臂 1 和 3 上的磁力方向相反，并且没有磁力施加在臂 2 或 4 上，因为 B 平行于这些臂中的电流流动方向。

如图 5-6(b) 所示，环路的底视图显示，力 F₁ 和 F₃ 产生绕原点 O 的扭矩，导致环路沿顺时针方向旋转。两个力的力臂均为 a/2，但 d₁ 和 d₃ 方向相反，因此总磁矩为

$$\begin{aligned}\mathbf{T} &= \mathbf{d}_1 \times \mathbf{F}_1 + \mathbf{d}_3 \times \mathbf{F}_3 \\ &= \left(-\hat{\mathbf{x}}\frac{a}{2}\right) \times (\hat{\mathbf{z}}IbB_0) + \left(\hat{\mathbf{x}}\frac{a}{2}\right) \times (-\hat{\mathbf{z}}IbB_0) \\ &= \hat{\mathbf{y}}IabB_0 = \hat{\mathbf{y}}IAB_0,\end{aligned} \tag{5.16}$$

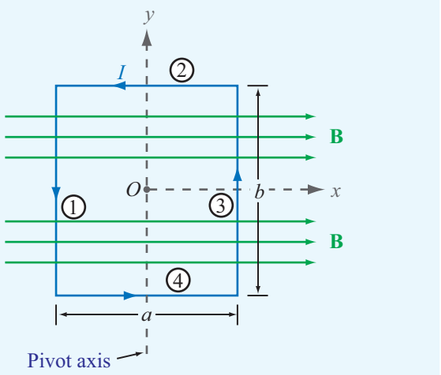


Figure 5-6 Rectangular loop pivoted along the *y* axis: (a) front view and (b) bottom view. The combination of forces $\mathbf{F}_1$ and $\mathbf{F}_3$ on the loop generates a torque that tends to rotate the loop in a clockwise direction as shown in (b).

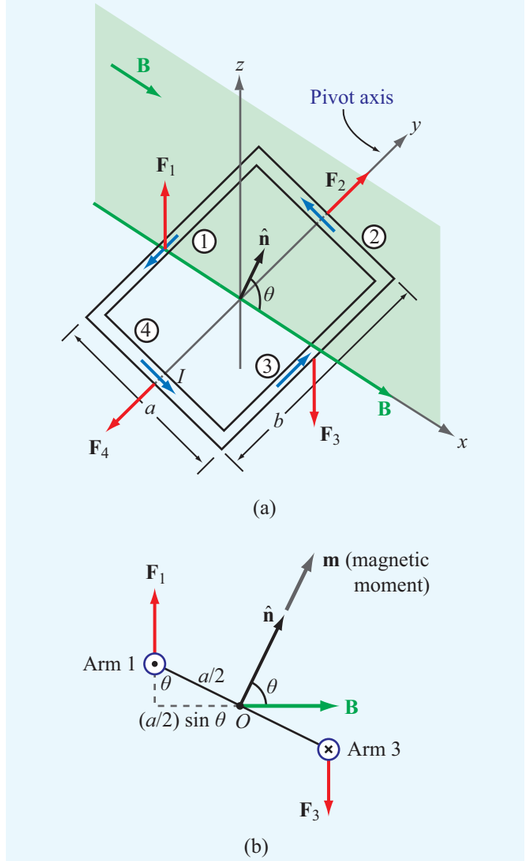


Figure 5-7 Rectangular loop in a uniform magnetic field with flux density **B** whose direction is perpendicular to the rotation axis of the loop but makes an angle θ with the loop's surface normal $\hat{\mathbf{n}}$.

where $A = ab$ is the area of the loop. The right-hand rule tells us that the sense of rotation is clockwise. The result given by Eq. (5.16) is valid only when the magnetic field **B** is parallel to the plane of the loop. As soon as the loop starts to rotate, the torque **T** decreases, and at the end of one quarter of a complete rotation, the torque becomes zero, as discussed next.

Magnetic Field Perpendicular to the Axis of a Rectangular Loop

For the situation represented by **Fig. 5-7**, where $\mathbf{B} = \hat{\mathbf{x}}B_0$, the field is still perpendicular to the loop's axis of rotation, but because its direction may be at any angle θ with respect to the loop's surface normal $\hat{\mathbf{n}}$, we may now have nonzero forces on all four arms of the rectangular loop. However, forces $\mathbf{F}_2$ and $\mathbf{F}_4$ are equal in magnitude and opposite in direction and are along the rotation axis; hence, the net torque contributed by their combination is zero. The directions of the currents in arms 1 and 3 are always perpendicular to **B** regardless of the magnitude of θ. Hence, $\mathbf{F}_1$ and $\mathbf{F}_3$ have the same expressions given previously by Eqs. (5.15a,b), and for $0 \le \theta \le \pi/2$, their moment arms are of magnitude $(a/2)\sin\theta$, as illustrated in **Fig. 5-7(b)**. Consequently, the magnitude of the net torque

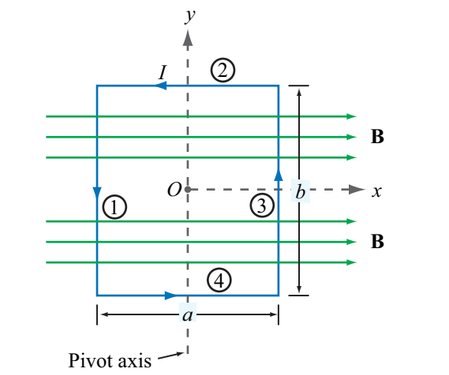


图 5-6 沿 y 轴旋转的矩形环：(a) 前视图和 (b) 底视图。环上的力 F_1 和 F_3 的组合会产生一个扭矩，该扭矩倾向于使环沿顺时针方向旋转，如 (b) 所示。

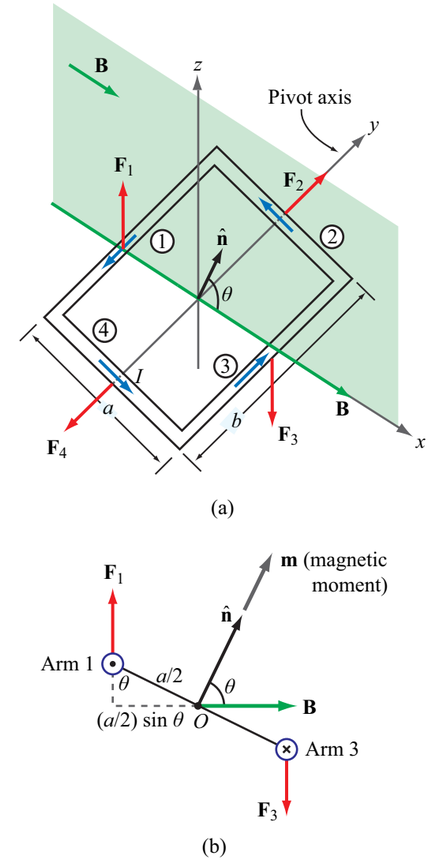


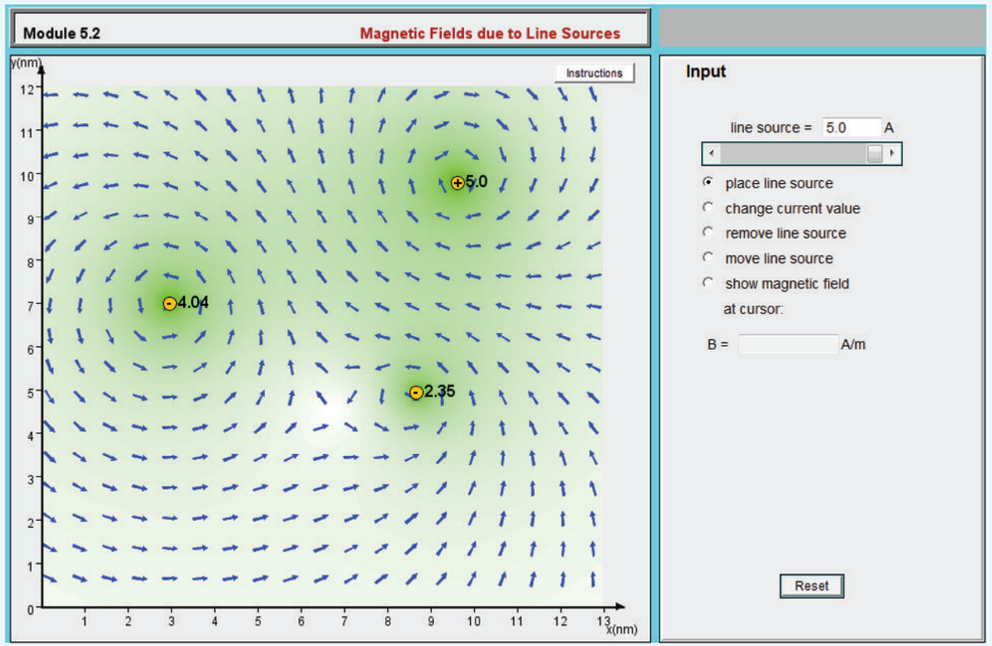
图 5-7 磁通密度为 B 的均匀磁场中的矩形环路，其方向垂直于环路的旋转轴，但与环路表面法线 n̂ 成角度 θ。

其中 A = ab 是环的面积。右手定则告诉我们，旋转方向是顺时针方向。方程给出的结果。 (5.16) 仅当磁场 B 平行于环路平面时才有效。一旦环路开始旋转，扭矩 T 就会减小，并且在四分之一完整旋转结束时，扭矩变为零，如下所述。

垂直于矩形环轴的磁场

对于图 5-7 所示的情况，其中 $B = \hat{x}B_0$，场仍然垂直于环的旋转轴，但由于其方向可能相对于环的表面法线 n̂ 成任何角度 θ，因此我们现在可以在矩形环的所有四个臂上施加非零力。然而，力 F_2 和 F_4 大小相等，方向相反，沿旋转轴；因此，它们的组合贡献的净扭矩为零。无论 θ 的大小如何，臂 1 和 3 中的电流方向始终垂直于 B。因此，F_1 和 F_3 具有与之前方程 1 给出的相同表达式。 (5.15a,b)，对于 $0 \le \theta \le \pi/2$，它们的力臂大小为 $(a/2)\sin\theta$，如图 5-7(b) 所示。因此，净扭矩的大小

Module 5.2 Magnetic Fields Due to Line Sources You can place z-directed linear currents anywhere in the display plane (x–y plane), select their magnitudes and directions, and then observe the spatial pattern of the induced magnetic flux $\mathbf{B}(x, y)$.



exerted by the magnetic field about the axis of rotation is the same as that given by Eq. (5.16), but modified by $\sin\theta$:

$$T = IAB_0 \sin\theta. \tag{5.17}$$

According to Eq. (5.17), the torque is maximum when the magnetic field is parallel to the plane of the loop ($\theta = 90°$) and zero when the field is perpendicular to the plane of the loop ($\theta = 0$). If the loop consists of N turns with each contributing a torque given by Eq. (5.17), then the total torque is

$$T = NIAB_0 \sin\theta. \tag{5.18}$$

The quantity NIA is called the ***magnetic moment*** m of the loop. Now, consider the vector

$$\mathbf{m} = \hat{\mathbf{n}} NIA = \hat{\mathbf{n}} m \qquad (\text{A}\cdot\text{m}^2), \tag{5.19}$$

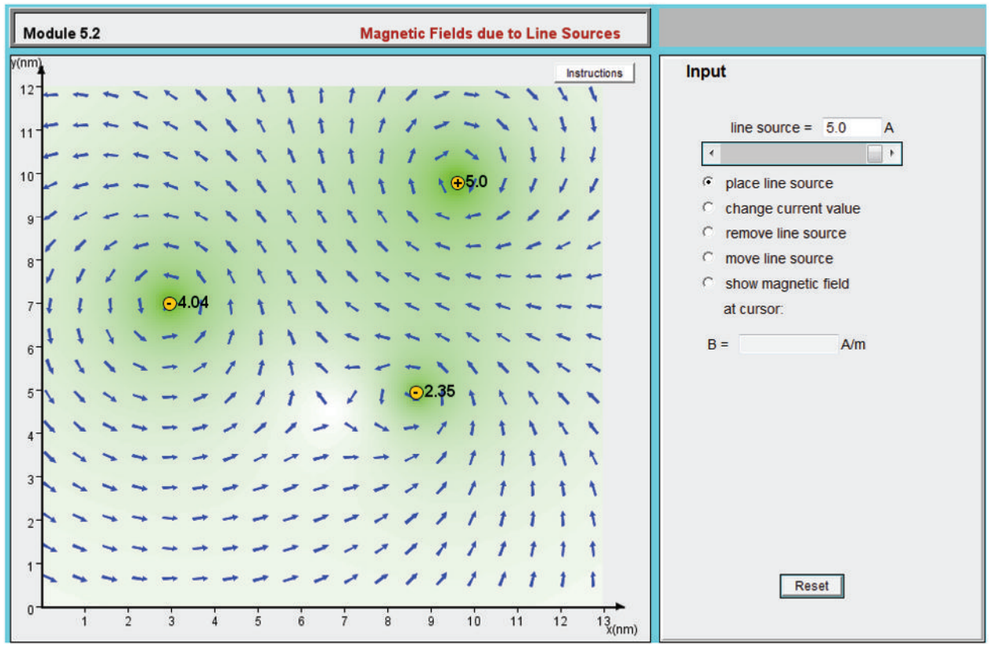
where $\hat{\mathbf{n}}$ is the surface normal of the loop and governed by the following ***right-hand rule:*** *When the four fingers of the right hand advance in the direction of the current I, the direction of the thumb specifies the direction of* $\hat{\mathbf{n}}$. In terms of $\mathbf{m}$, the torque vector $\mathbf{T}$ can be written as

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \qquad (\text{N}\cdot\text{m}). \tag{5.20}$$

Even though the derivation leading to Eq. (5.20) was obtained for $\mathbf{B}$ perpendicular to the axis of rotation of a rectangular loop, the expression is valid for any orientation of $\mathbf{B}$ and for a loop of any shape.

Concept Question 5-3: How is the direction of the magnetic moment of a loop defined?

模块 5.2 线源产生的磁场 您可以将 z 向线性电流放置在显示平面（x-y 平面）的任何位置，选择它们的大小和方向，然后观察感应磁通量 B(x, y) 的空间模式。



围绕旋转轴施加的磁场与式（1）给出的相同。 (5.16)，但由 sin θ 修改：

$$T = IAB_0 \sin\theta. \tag{5.17}$$

根据方程。根据方程（5.17），当磁场平行于环路平面（$\theta = 90°$）时，扭矩最大；当磁场垂直于环路平面（$\theta = 0$）时，扭矩为零。如果环路由 N 匝组成，每匝都贡献由方程式给出的扭矩。 (5.17) 则总扭矩为

$$T = NIAB_0 \sin\theta. \tag{5.18}$$

量 NIA 称为环路的磁矩 m。现在，考虑向量

$$\mathbf{m} = \hat{\mathbf{n}} NIA = \hat{\mathbf{n}} m \qquad (\text{A}\cdot\text{m}^2), \tag{5.19}$$

其中，n̂ 是环的表面法线，并受以下右手定则约束：当右手的四个手指沿电流 I 的方向前进时，拇指的方向指定 n̂ 的方向。用 m 表示，扭矩矢量 T 可以写为

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \qquad (\text{N}\cdot\text{m}). \tag{5.20}$$

即使推导得出方程。 (5.20) 是针对垂直于矩形环的旋转轴的 B 获得的，该表达式对于 B 的任何方向和任何形状的环都有效。

概念问题 5-3：如何定义环路磁矩的方向？